

24.0.1 UGC NET CSE | December 2012 | Part 3 | Question: 60



The transform which possesses 'multi-resolution' property is

- A. Fourier transform
 B. Short-time Fourier transform
 C. Wavelet transform
 D. Karhunen-Loere transform

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Answer key

24.1

Butterworth Lowpass Filter (1)

24.1.1 Butterworth Lowpass Filter: UGC NET CSE | December 2014 | Part 3 | Question: 66



A Butterworth lowpass filter of order n , with a cutoff frequency at distance D_0 from the origin, has the transfer function $H(u, v)$ given by

- A. $\frac{1}{1 + \left[\frac{D(u,v)}{D_0} \right]^{2n}}$
 B. $\frac{1}{1 + \left[\frac{D(u,v)}{D_0} \right]^n}$
 C. $\frac{1}{1 + \left[\frac{D_0}{D(u,v)} \right]^{2n}}$
 D. $\frac{1}{1 + \left[\frac{D_0}{D(u,v)} \right]^n}$

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Answer key

24.2

Digital Signal Processing (1)

24.2.1 Digital Signal Processing: UGC NET CSE | June 2012 | Part 2 | Question: 18



Page Shift Keying (PSK) Method is used to modulate digital signal at 9600 bps using 16 level. Find the line signals and speed (i.e., modulation rate)

- A. 2400 bauds
 B. 1200 bauds
 C. 4800 bauds
 D. 9600 bauds

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Answer key

24.3

Discrete Memoryless Channel (1)

24.3.1 Discrete Memoryless Channel: UGC NET CSE | July 2016 | Part 3 | Question: 58



Consider a discrete memoryless channel and assume that $H(x)$ is the amount of information per symbol at the input of the channel; $H(y)$ is the amount of information per symbol at the output of the channel. $H(x | y)$ is the amount of uncertainty remaining on x knowing y ; and $I(x; y)$ is the information transmission. Which of the following does not define the channel capacity of a discrete memoryless channel?

- A. $\max I(x; y); p(x)$
 B. $\max [H(y) - H(y | x)]; p(x)$
 C. $\max [H(x) - H(x | y)]; p(x)$
 D. $\max H(x | y); p(x)$

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Answer key

24.4

Spatial Mask (1)

24.4.1 Spatial Mask: UGC NET CSE | December 2014 | Part 3 | Question: 64



Given two spatial masks

$$S_1 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

$$S_2 = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -8 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

The Laplacian of an image at all points (x,y) can be implemented by convolving the image with spatial mask. Which of the following can be used as the spatial mask ?

- A. only S_1
- B. only S_2
- C. Both S_1 and S_2
- D. None of these

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Answer key 

Answer Keys

24.0.1	C	24.1.1	A	24.2.1	A	24.3.1	D	24.4.1	B
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